

Graph–Sequence Deep Models for Kinase Drug–Target Affinity: Verified Benchmarks, Two Failure Modes, and Their Fixes

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Abstract

We build and verify graph-neural-network + sequence-CNN models for drug–target affinity on the DAVIS kinase benchmark, quantify two failure modes (affinity range compression; point-mutation blindness), and evaluate two fixes (affinity-weighted loss; attention pooling) with logged, committed evidence.

1 Problem and data

DAVIS: 68 drugs $\times$ 442 kinases, affinities as K_d in nM. We model

$$\text{p}K_d = -\log_{10}(K_d/10^9), \quad K_d = 0 \Rightarrow K_d := 10^5 \text{ nM} \quad (1)$$

following the DeepDTA convention.

2 Models

2.1 Molecular graph encoder

Each drug is a graph $G = (V, E)$ with atom features $x_v \in \mathbb{R}^F$. With $\tilde{A} = A + I$ and degree matrix

$$D_{ii} = \sum_j \tilde{A}_{ij}, \quad \hat{A} = D^{-1/2} \tilde{A} D^{-1/2}, \quad (2)$$

graph convolution with normalized adjacency $\hat{A}$:

$$H^{(l+1)} = \phi(\hat{A} H^{(l)} W^{(l)}), \quad z_{\text{lig}} = \text{meanpool}(H^{(L)}). \quad (3)$$

2.2 Protein sequence encoders

Baseline CNN over residue one-hots $S \in \{0, 1\}^{L \times 21}$:

$$h^{(l+1)} = \phi(W^{(l)} * h^{(l)}), \quad z_{\text{prot}} = \frac{1}{L} \sum_i h_i^{(L)}. \quad (4)$$

Attention-pooled variant (fix for mutation blindness), with mask $m_i = \mathbb{I}[\text{residue } i \text{ real}]$:

$$\alpha = \text{softmax}((U h_i) + \log m_i), \quad z_{\text{prot}}^{\text{attn}} = \sum_{i=1}^L \alpha_i V h_i, \quad \sum_i \alpha_i = 1, \quad \alpha_i = 0 \quad \forall i \in \text{pad}. \quad (5)$$

2.3 Affinity head and losses

$$\hat{y} = w^\top [z_{\text{lig}}; z_{\text{prot}}] + b. \quad (6)$$

Plain training minimizes $\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$. The affinity-weighted loss (fix for range compression) upweights strong binders:

$$\mathcal{L}_w = \frac{1}{n} \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2, \quad w_i = \begin{cases} 6 & y_i \geq 7 \\ 1 & \text{otherwise} \end{cases}. \quad (7)$$

3 Metrics

Ranking quality by the concordance index over held-out pairs:

$$C = \frac{\sum_{i < j} \mathbb{I}[y_i > y_j] (\mathbb{I}[\hat{y}_i > \hat{y}_j] + \frac{1}{2} \mathbb{I}[\hat{y}_i = \hat{y}_j])}{\sum_{i < j} \mathbb{I}[y_i \neq y_j]}. \quad (8)$$

Calibration on strong binders ($y \geq 7$): we report $\bar{y}_{\text{strong}}$, $\bar{\hat{y}}_{\text{strong}}$, and $\text{MSE}_{\text{strong}}$ on the 417 held-out strong pairs. Gradient steps use Adam:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad \hat{m}_t = m_t / (1 - \beta_1^t), \quad \theta_t = \theta_{t-1} - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon). \quad (9)$$

4 Failure modes (verified)

A. Range compression. The MSE-optimal predictor is the conditional mean,

$$\hat{y}^*(x) = \mathbb{E}[y | x], \quad \text{Var}(\hat{y}^*) \leq \text{Var}(y), \quad (10)$$

so whenever the model cannot separate strong from weak binders its predictions contract toward the global mean. With $\approx 95\%$ of training pairs weak ($\text{p}K_d \approx 5$), predictions collapse toward the weak affinity: strong-binder $\hat{y} = 6.11$ vs $\bar{y} = 7.90$. Formally, writing the strong-binder residual

$$\text{MSE}_{\text{strong}} = \frac{1}{|S|} \sum_{i \in S} (y_i - \hat{y}_i)^2, \quad S = \{i : y_i \geq 7\}, \quad |S| = 417, \quad (11)$$

the compressed model pays $\approx (7.90 - 6.11)^2 \approx 3.2$ squared $\text{p}K_d$ units of pure bias before any variance term.

B. Mutation blindness. Truncation to $L = 600$ plus global mean pooling gives bit-identical predictions for all 8 KIT point mutants ($\Delta z = 0$ exactly, verified in tests).

5 Results

All numbers are read from committed files: `results/davis_log.jsonl`, `results/davis_log_attn.jsonl`, `results/strong_eval_log.jsonl`. Current state: baseline track test $C = 0.7048$ (epoch 45), attention track $C = 0.6575$ (epoch 18, still converging); weighted loss yields only unstable strong-binder recovery (series 6.0–6.6 vs 6.11 baseline). Negative results are kept.

6 Tool

`python -m targetscan ships screen` (any SMILES vs 442 kinases), `eval` (verified numbers), `verify` (ChEMBL corroboration).